

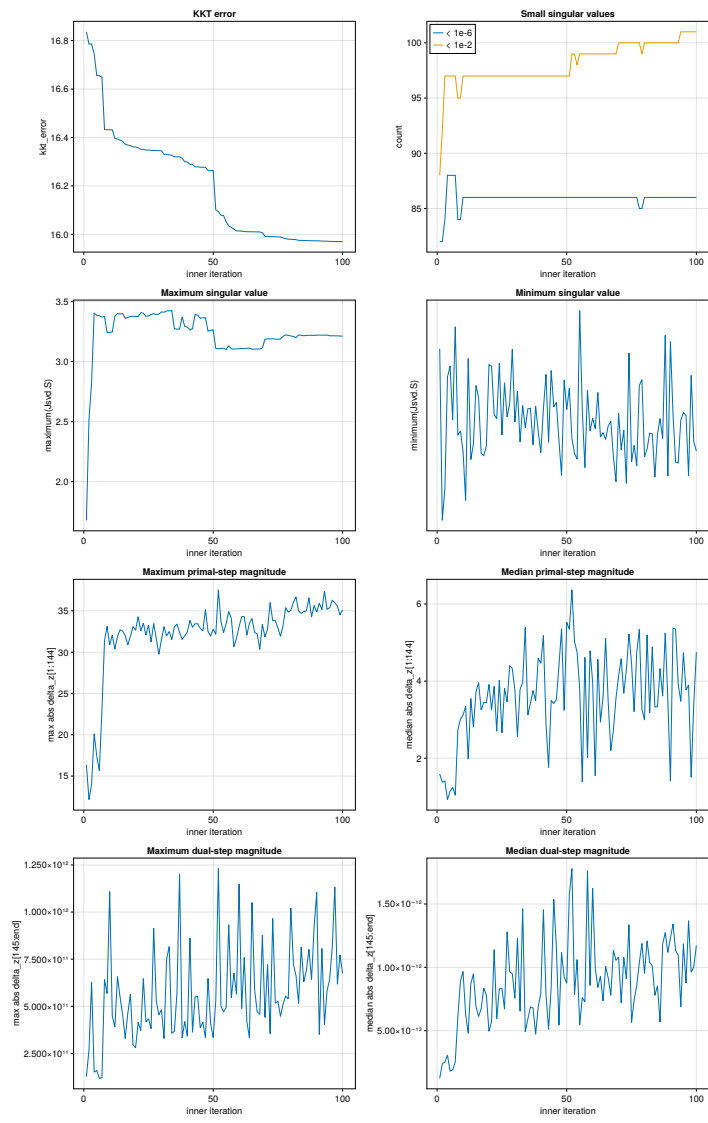
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
kkt_error	1.597008e+01	1.683624e+01	1.619596e+01	1.618192e+01
count(Jsvd.S < 1e-6)	8.200000e+01	8.800000e+01	8.592000e+01	8.600000e+01
count(Jsvd.S < 1e-2)	8.800000e+01	1.010000e+02	9.816000e+01	9.700000e+01
maximum(Jsvd.S)	1.677692e+00	3.425130e+00	3.232453e+00	3.220744e+00
minimum(Jsvd.S)	2.986024e-17	8.396667e-17	5.546353e-17	5.400809e-17
maximum(abs.(delta_z[1:144]))	1.215507e+01	3.754690e+01	3.224910e+01	3.308847e+01
median(abs.(delta_z[1:144]))	9.260410e-01	6.368560e+00	3.604847e+00	3.567126e+00
maximum(abs.(delta_z[145:end]))	1.172504e+11	1.230960e+12	5.712513e+11	5.290355e+11
median(abs.(delta_z[145:end]))	1.234450e-13	1.780073e-12	8.854307e-13	8.633885e-13

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\text{med} \delta z_{1:144} $	$\max \delta z_{145:} $	$\text{med} \delta z_{145:} $
1	1.683624e+01	82	88	1.677692e+00	7.413368e-17	1.635917e+01	1.590470e+00	1.260088e+11	1.234450e-13
2	1.678591e+01	82	92	2.520494e+00	2.986024e-17	1.215507e+01	1.380033e+00	2.674753e+11	2.378164e-13
3	1.678591e+01	84	97	2.821233e+00	3.802944e-17	1.399427e+01	1.405146e+00	6.269700e+11	2.506766e-13
4	1.674471e+01	88	97	3.404230e+00	6.682588e-17	2.012417e+01	9.260410e-01	1.501676e+11	3.071350e-13
5	1.665608e+01	88	97	3.386006e+00	6.965280e-17	1.743490e+01	1.139665e+00	1.590376e+11	1.809782e-13
6	1.665512e+01	88	97	3.381653e+00	5.585761e-17	1.565083e+01	1.246836e+00	1.172504e+11	1.897378e-13
7	1.664886e+01	88	97	3.371112e+00	7.979311e-17	2.293059e+01	1.038570e+00	1.204212e+11	2.540141e-13
8	1.643303e+01	84	95	3.375305e+00	5.182869e-17	3.138588e+01	2.737555e+00	6.414593e+11	6.077820e-13
9	1.643214e+01	84	95	3.242425e+00	5.294245e-17	3.313302e+01	3.012990e+00	5.675264e+11	8.917241e-13
10	1.643208e+01	86	97	3.243103e+00	4.764477e-17	3.086823e+01	3.116859e+00	1.108865e+12	9.668674e-13
11	1.643165e+01	86	97	3.246890e+00	3.499830e-17	3.207340e+01	3.346440e+00	4.480593e+11	6.285522e-13
12	1.639607e+01	86	97	3.379776e+00	7.158928e-17	3.036351e+01	1.982174e+00	3.900000e+11	4.801054e-13
13	1.639374e+01	86	97	3.399563e+00	4.553381e-17	3.189830e+01	3.550058e+00	6.575337e+11	8.695132e-13
14	1.638930e+01	86	97	3.397621e+00	4.948835e-17	3.269967e+01	2.802785e+00	5.498220e+11	9.484842e-13
15	1.638393e+01	86	97	3.400115e+00	6.464033e-17	3.255240e+01	3.722427e+00	4.552277e+11	6.886897e-13
16	1.637234e+01	86	97	3.359730e+00	6.152957e-17	3.199403e+01	3.961012e+00	3.284536e+11	6.123873e-13
17	1.636843e+01	86	97	3.369848e+00	4.710655e-17	3.086110e+01	3.248443e+00	4.627380e+11	6.733054e-13
18	1.636592e+01	86	97	3.376254e+00	4.655714e-17	3.190949e+01	3.447215e+00	5.645012e+11	8.351126e-13
19	1.636124e+01	86	97	3.376762e+00	4.905033e-17	3.309575e+01	3.444346e+00	2.959427e+11	7.814688e-13
20	1.636026e+01	86	97	3.376788e+00	7.002462e-17	3.265100e+01	3.916148e+00	2.790704e+11	4.947114e-13
21	1.635776e+01	86	97	3.374920e+00	6.960774e-17	3.429234e+01	3.254881e+00	4.146454e+11	5.715124e-13
22	1.635157e+01	86	97	3.406874e+00	5.719827e-17	3.256444e+01	3.863342e+00	3.711724e+11	1.139685e-12
23	1.635063e+01	86	97	3.402836e+00	5.599249e-17	3.352016e+01	2.698598e+00	6.470623e+11	5.939675e-13
24	1.634802e+01	86	97	3.376674e+00	7.034352e-17	3.206646e+01	4.026855e+00	4.174597e+11	8.298860e-13
25	1.634799e+01	86	97	3.380572e+00	5.186384e-17	3.329594e+01	2.657143e+00	4.335682e+11	8.325393e-13
26	1.634718e+01	86	97	3.392409e+00	6.532189e-17	3.124464e+01	3.810417e+00	3.804456e+11	6.701646e-13
27	1.634621e+01	86	97	3.399795e+00	5.650168e-17	3.346452e+01	3.460599e+00	9.138893e+11	1.281084e-12
28	1.634616e+01	86	97	3.391660e+00	6.009741e-17	3.164673e+01	4.403435e+00	5.282194e+11	9.651131e-13
29	1.634583e+01	86	97	3.395826e+00	7.396111e-17	2.974323e+01	4.323571e+00	4.543889e+11	9.506243e-13
30	1.634472e+01	86	97	3.413962e+00	5.521106e-17	3.151796e+01	3.763003e+00	4.816355e+11	7.554396e-13
31	1.633036e+01	86	97	3.411524e+00	6.320419e-17	3.309411e+01	2.561585e+00	3.282889e+11	1.232613e-12
32	1.632952e+01	86	97	3.421862e+00	5.007974e-17	3.196967e+01	3.787535e+00	7.521921e+11	6.536894e-13
33	1.632862e+01	86	97	3.422571e+00	5.946831e-17	3.249270e+01	3.966719e+00	8.169227e+11	1.463642e-12
34	1.632602e+01	86	97	3.425130e+00	5.376691e-17	3.152678e+01	5.398980e+00	3.575521e+11	4.898276e-13
35	1.631993e+01	86	97	3.272920e+00	5.863210e-17	3.308283e+01	3.117878e+00	3.659712e+11	5.912965e-13
36	1.631990e+01	86	97	3.270267e+00	5.874615e-17	3.337652e+01	3.429595e+00	5.722932e+11	6.870457e-13
37	1.631985e+01	86	97	3.270297e+00	4.935712e-17	3.240365e+01	3.756152e+00	1.202871e+12	6.790821e-13
38	1.631489e+01	86	97	3.371788e+00	5.934509e-17	3.153532e+01	3.493606e+00	3.311462e+11	4.710932e-13
39	1.629952e+01	86	97	3.292743e+00	5.279620e-17	3.198734e+01	4.588567e+00	4.191012e+11	6.891740e-13
40	1.629918e+01	86	97	3.287319e+00	4.726948e-17	3.244289e+01	4.462916e+00	3.405753e+11	7.947241e-13
41	1.628907e+01	86	97	3.263576e+00	5.861370e-17	3.385101e+01	5.181657e+00	8.624743e+11	1.455529e-12
42	1.628894e+01	86	97	3.275442e+00	6.748506e-17	3.301007e+01	2.891372e+00	3.618780e+11	7.842310e-13
43	1.627883e+01	86	97	3.392059e+00	4.999278e-17	3.343382e+01	1.761049e+00	5.502675e+11	5.102321e-13
44	1.627859e+01	86	97	3.386773e+00	6.850489e-17	3.344045e+01	3.496503e+00	5.527511e+11	8.439275e-13
45	1.627713e+01	86	97	3.361493e+00	5.914063e-17	3.287496e+01	3.421343e+00	3.850193e+11	1.537742e-12
46	1.627701e+01	86	97	3.365031e+00	6.026613e-17	3.257584e+01	3.519231e+00	4.154672e+11	1.156794e-12
47	1.627691e+01	86	97	3.365133e+00	5.035104e-17	3.514923e+01	4.399846e+00	3.314254e+11	5.435759e-13
48	1.626392e+01	86	97	3.254158e+00	4.147165e-17	3.248765e+01	5.356937e+00	6.475294e+11	1.116985e-12
49	1.626383e+01	86	97	3.260551e+00	6.601327e-17	3.192723e+01	3.241323e+00	4.063247e+11	9.255274e-13
50	1.626336e+01	86	97	3.263591e+00	5.843369e-17	3.280033e+01	5.525212e+00	3.335446e+11	8.739334e-13
51	1.610047e+01	86	97	3.109430e+00	6.395809e-17	3.215465e+01	5.345547e+00	5.298515e+11	1.572290e-12
52	1.609348e+01	86	99	3.106940e+00	5.101976e-17	3.754690e+01	6.368560e+00	1.230960e+12	1.780073e-12
53	1.607870e+01	86	99	3.110625e+00	4.710306e-17	3.368155e+01	5.028150e+00	5.007743e+11	7.856185e-13
54	1.607823e+01	86	98	3.111039e+00	4.562668e-17	3.236520e+01	4.741804e+00	4.697033e+11	1.061655e-12
55	1.605193e+01	86	99	3.100745e+00	8.396667e-17	3.345145e+01	3.805404e+00	4.935189e+11	5.444423e-13
56	1.603446e+01	86	99	3.131265e+00	5.843134e-17	3.491363e+01	1.382680e+00	9.330259e+11	7.611718e-13
57	1.602866e+01	86	99	3.107164e+00	4.345967e-17	3.407120e+01	4.616648e+00	5.444557e+11	7.264285e-13
58	1.602132e+01	86	99	3.103330e+00	6.343241e-17	3.066026e+01	2.010648e+00	6.754993e+11	1.763041e-12
59	1.601416e+01	86	99	3.105870e+00	5.296117e-17	3.168058e+01	4.787271e+00	5.635784e+11	8.567024e-13
60	1.601404e+01	86	99	3.106598e+00	5.621717e-17	3.310050e+01	3.972755e+00	1.149230e+12	1.625814e-12
61	1.601353e+01	86	99	3.110043e+00	5.350811e-17	3.430091e+01	1.544414e+00	4.879782e+11	9.935549e-13
62	1.601160e+01	86	99	3.107471e+00	6.283857e-17	3.429627e+01	4.567153e+00	7.592798e+11	8.408376e-13

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\text{med} \delta z_{1:144} $	$\max \delta z_{145:} $	$\text{med} \delta z_{145:} $
63	1.601087e+01	86	99	3.110620e+00	5.132693e-17	3.205176e+01	2.933247e+00	4.184189e+11	9.295314e-13
64	1.601063e+01	86	99	3.111998e+00	5.249330e-17	3.349480e+01	3.574112e+00	3.323852e+11	7.353479e-13
65	1.601062e+01	86	99	3.103607e+00	5.074648e-17	3.406812e+01	5.113576e+00	1.050983e+12	1.008130e-12
66	1.601032e+01	86	99	3.103076e+00	5.424926e-17	3.239499e+01	3.444920e+00	5.890446e+11	9.162041e-13
67	1.601018e+01	86	99	3.104722e+00	5.545078e-17	3.221515e+01	2.195730e+00	4.706627e+11	7.789103e-13
68	1.601005e+01	86	99	3.104232e+00	4.622306e-17	3.033790e+01	2.732269e+00	4.571083e+11	1.133974e-12
69	1.600684e+01	86	99	3.113985e+00	3.980997e-17	3.337914e+01	3.560139e+00	8.785799e+11	1.055349e-12
70	1.599118e+01	86	100	3.186005e+00	5.748260e-17	3.184604e+01	4.125930e+00	4.405357e+11	1.081541e-12
71	1.599102e+01	86	100	3.189124e+00	4.807768e-17	3.276486e+01	4.583980e+00	7.210931e+11	7.203712e-13
72	1.599079e+01	86	100	3.189220e+00	5.314366e-17	3.600787e+01	3.682038e+00	3.542452e+11	1.079238e-12
73	1.599026e+01	86	100	3.190512e+00	3.938120e-17	3.381293e+01	4.282394e+00	9.659976e+11	9.086862e-13
74	1.598995e+01	86	100	3.187125e+00	7.301890e-17	3.381379e+01	5.224636e+00	5.171108e+11	1.337076e-12
75	1.598985e+01	86	100	3.185842e+00	4.489615e-17	3.291898e+01	4.462495e+00	5.270020e+11	5.619220e-13
76	1.598936e+01	86	100	3.188464e+00	4.919147e-17	3.191616e+01	3.207684e+00	4.463432e+11	7.424055e-13
77	1.598537e+01	86	100	3.210496e+00	4.393584e-17	3.320610e+01	4.742210e+00	5.115327e+11	8.572638e-13
78	1.598195e+01	85	100	3.222282e+00	6.466824e-17	3.539406e+01	5.348081e+00	5.526928e+11	1.036820e-12
79	1.598001e+01	85	99	3.218508e+00	6.614813e-17	3.484629e+01	3.272057e+00	5.401205e+11	1.188745e-12
80	1.597964e+01	86	100	3.213556e+00	4.623915e-17	3.506054e+01	2.989171e+00	1.021118e+12	9.523770e-13
81	1.597891e+01	86	100	3.209076e+00	4.831555e-17	3.604717e+01	5.196431e+00	7.223466e+11	1.209529e-12
82	1.597807e+01	86	100	3.200262e+00	5.233887e-17	3.668683e+01	3.174837e+00	6.622509e+11	1.038445e-12
83	1.597518e+01	86	100	3.222038e+00	5.214609e-17	3.498962e+01	4.889606e+00	5.150750e+11	1.010824e-12
84	1.597509e+01	86	100	3.219341e+00	4.101850e-17	3.469569e+01	3.328910e+00	8.148324e+11	7.795356e-13
85	1.597468e+01	86	100	3.215164e+00	5.207262e-17	3.490635e+01	3.334625e+00	6.302998e+11	8.538257e-13
86	1.597413e+01	86	100	3.218653e+00	5.608785e-17	3.499778e+01	4.326456e+00	6.901772e+11	5.672258e-13
87	1.597376e+01	86	100	3.218848e+00	5.098570e-17	3.657905e+01	3.608455e+00	8.017110e+11	1.183041e-12
88	1.597365e+01	86	100	3.219018e+00	7.757207e-17	3.429766e+01	5.251991e+00	6.415881e+11	1.274883e-12
89	1.597343e+01	86	100	3.218415e+00	4.128908e-17	3.564128e+01	3.509465e+00	9.219127e+11	1.117983e-12
90	1.597324e+01	86	100	3.219796e+00	7.596207e-17	3.485705e+01	1.404226e+00	1.104996e+12	1.232619e-12
91	1.597322e+01	86	100	3.221146e+00	5.710442e-17	3.591948e+01	5.375588e+00	3.502341e+11	1.342044e-12
92	1.597192e+01	86	100	3.220238e+00	4.485974e-17	3.514776e+01	5.344428e+00	8.065208e+11	1.130590e-12
93	1.597184e+01	86	100	3.220150e+00	4.467593e-17	3.739950e+01	3.920103e+00	4.026450e+11	1.096414e-12
94	1.597171e+01	86	101	3.220341e+00	5.567696e-17	3.519489e+01	3.472444e+00	5.836285e+11	6.889625e-13
95	1.597086e+01	86	101	3.214967e+00	5.767851e-17	3.537326e+01	4.737457e+00	6.416599e+11	1.186159e-12
96	1.597076e+01	86	101	3.214689e+00	5.687443e-17	3.627773e+01	3.767188e+00	8.138244e+11	8.764876e-13
97	1.597064e+01	86	101	3.214566e+00	4.137306e-17	3.599344e+01	3.899052e+00	1.133275e+12	1.368469e-12
98	1.597036e+01	86	101	3.214204e+00	6.723891e-17	3.556248e+01	1.504559e+00	6.167878e+11	9.613336e-13
99	1.597021e+01	86	101	3.213333e+00	5.020365e-17	3.450915e+01	3.405934e+00	7.714133e+11	1.003863e-12
100	1.597008e+01	86	101	3.210758e+00	4.774935e-17	3.512733e+01	4.752178e+00	6.745121e+11	1.172124e-12